TEXORA · LECTURE SHEET

Carboxylic Acids

The –COOH group: resonance and acidity, substituent effects on strength, preparations, and the characteristic reactions — salts and esters, acid derivatives, decarboxylation, the Hell–Volhard–Zelinsky reaction, and formic acid's double life as a reducing agent.

CHE-1205 · CHEMISTRY-II · TOPIC 4 · LEVEL 1 TERM 2

1. Structure and acidity

The **carboxyl group** –COOH joins a carbonyl and a hydroxyl on one carbon. Its acidity (pKa ≈ 4–5, a million times stronger than phenol) comes from the **carboxylate anion**: $RCOO^-$ spreads its charge over two equivalent oxygens (both C–O bonds equal, 1.27 Å — resonance caught in the act by X-ray measurements), while the undissociated acid is far less stabilised. Everything in this topic is a variation on how substituents treat that anion:

ACID	PKA	LESSON
Trichloroacetic acid	0.65	Electron-withdrawing groups stabilise $RCOO^-$ → stronger acid; more of them, and closer, the stronger
Chloroacetic acid	2.87	
Formic acid (HCOOH)	3.75	
Benzoic acid	4.20	Slightly stronger than acetic — the ring carbon is sp^2 and mildly withdrawing
Acetic acid	4.76	Reference point
Propanoic acid	4.87	The extra electron-donating CH_3 destabilises the anion slightly → weaker

Position matters as much as identity: α-chloropropanoic acid (pKa 2.8) is far stronger than the β-chloro isomer (pKa ≈ 4.0) — the inductive effect fades with distance.

2. Preparation

- **Oxidation** of primary alcohols or aldehydes ($KMnO_4$, $K_2Cr_2O_7$) — the aldehyde is the halfway house (Topic 3).

- **Hydrolysis** of nitriles (acid or alkaline), esters, amides and acyl chlorides.
- **Grignard + CO_2:** R–MgX poured onto dry ice gives, after work-up, R–COOH with one extra carbon — the clean way to lengthen a chain into an acid.
- **Kolbe–Schmitt** for salicylic acid (phenol route, Topic 2); oxidation of alkylbenzenes ($KMnO_4$ takes any benzylic side chain down to –COOH: toluene → benzoic acid).

3. Reactions

REACTION	OUTCOME / SIGNIFICANCE
With $NaHCO_3$ or Na_2CO_3	Brisk effervescence of CO_2 — the defining test: acids stronger than carbonic acid pass; phenols (Topic 2) fail
Fischer esterification	RCOOH + R′OH ⇌ RCOOR′ + H_2O (conc. H_2SO_4); an equilibrium — drive it with excess alcohol or water removal
To acid derivatives	$SOCl_2$ or PCl_5 → acyl chloride; NH_3 then heat → amide; two molecules − H_2O (or P_2O_5) → anhydride. All derivatives hydrolyse back to the acid, fastest for the chloride
Reduction ($LiAlH_4$)	–COOH → –CH_2OH (primary alcohol); $NaBH_4$ is too weak — a useful selectivity note
Decarboxylation	Soda-lime (NaOH/CaO, heat) removes –COOH as carbonate: RCOONa → R–H, an alkane with one carbon fewer. β-keto acids and malonic acid decarboxylate on simple heating. Kolbe electrolysis of carboxylate salts couples two alkyl radicals: 2 $RCOO^-$ → R–R + 2 CO_2
Hell–Volhard–Zelinsky	Br_2/red P brominates the α-carbon (through the acyl bromide's enol): propanoic acid → 2-bromopropanoic acid. Needs an α-hydrogen — benzoic and formic acids fail

Formic acid, the exception: H–COOH carries a hydrogen on the carbonyl carbon, making it aldehyde-like — it reduces Tollens' and Fehling's reagents and decolourises $KMnO_4$, the only carboxylic acid that does. It also dehydrates with conc. H_2SO_4 to CO + H_2O, a laboratory preparation of carbon monoxide.

4. Important members

Oxalic acid (HOOC–COOH) — the simplest dicarboxylic acid, a reducing agent and rust/ink-stain remover. **Malonic acid** loses CO_2 on heating (two –COOH on one carbon cannot survive heat), the basis of malonic-ester synthesis. **Lactic acid** — the hydroxy acid of Topic 1's optical isomerism.

Salicylic acid — hydroxy acid of aspirin fame, made by Kolbe–Schmitt. **Higher fatty acids** (stearic, oleic) and their sodium salts are soaps; their textile relevance is direct: scouring agents, softeners and dyeing auxiliaries are built from carboxylate chemistry.

5. Key points

- Acidity is carboxylate resonance stabilisation; EWG strengthen (more, closer = stronger), EDG weaken.
- $NaHCO_3$ effervescence separates carboxylic acids from phenols.
- Derivatives (chloride, amide, anhydride, ester) all hydrolyse back; $LiAlH_4$ reduces the acid to a 1° alcohol.
- Decarboxylation shortens the chain by one carbon (soda lime); HVZ brominates the α-position and needs an α-H.
- Formic acid alone among acids reduces Tollens' reagent — it is structurally half aldehyde.

6. Practice

Answer in your own words; brief model answers follow.

1. Arrange in decreasing acid strength: acetic acid, chloroacetic acid, trichloroacetic acid, propanoic acid — and justify the order briefly.
2. Two white solids are benzoic acid and phenol. Give one chemical test that distinguishes them, with the observation.
3. Propanoic acid is treated with Br_2/red phosphorus. Name the product and the reaction; state why benzoic acid gives no analogous product.
4. Sodium propanoate is heated with soda lime. Name the gas-producing salt's organic product, and the general name of this reaction type.
5. Formic acid gives a silver mirror with Tollens' reagent but acetic acid does not. Explain, using structure.
6. Starting from acetic acid, outline preparations of (a) acetyl chloride, (b) acetamide, (c) ethyl acetate, naming reagents.

Answer key

1. Trichloroacetic > chloroacetic > acetic > propanoic. Each chlorine withdraws electron density (−I), stabilising the carboxylate anion and easing proton loss; three chlorines beat one. Propanoic trails acetic because its extra methyl donates (+I), destabilising the anion.

2. Add aqueous $NaHCO_3$: benzoic acid effervesces briskly (CO_2 evolved); phenol, weaker than carbonic acid, shows no reaction. ($FeCl_3$ also works — phenol turns violet — but the bicarbonate test uses acidity itself.)
3. 2-Bromopropanoic acid — the Hell–Volhard–Zelinsky reaction. Benzoic acid has no α-carbon at all (the –COOH is bonded straight to the ring), so there is no α-hydrogen to substitute.
4. Ethane. Decarboxylation — the –COONa group leaves as sodium carbonate, and the alkyl residue takes a hydrogen: the chain shortens by exactly one carbon (propanoate, C_3 → ethane, C_2).
5. Formic acid is H–C(=O)–OH: the hydrogen sits directly on the carbonyl carbon, so the molecule contains an aldehyde-like –CHO arrangement and oxidises easily to CO_2, reducing Ag^+ to silver. In acetic acid that position carries a methyl group; the aldehyde character vanishes and Tollens' reagent is untouched.
6. (a) $SOCl_2$ (or PCl_5) → acetyl chloride. (b) NH_3 to form ammonium acetate, then strong heating → acetamide. (c) Ethanol with concentrated H_2SO_4 catalyst (Fischer esterification), removing water or using excess ethanol to drive the equilibrium → ethyl acetate.

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